Shuntaro Suzuki

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Ph.D. applicant in computational neuroscience, working on representation learning and foundation models for neural signals, with a focus on decoding the human cognitive processes and advancing brain-computer interfaces.

Education

Keio University **Apr. 2025 – Expected Mar. 2027**
Master of Information and Computer Science *Kanagawa, Japan*

Keio University **Apr. 2021 – Mar. 2025**
Bachelor of Information and Computer Science *Kanagawa, Japan*

Publications

Journal Articles

[1] **Shuntaro Suzuki**, Shunya Nagashima, Komei Sugiura. "Cortical-SSM: A Deep State Space Model for Motor Imagery Decoding from EEG Signals", *Journal of Neural Engineering*, vol. 23, no. 4, 046032, 2026. doi: 10.1088/1741-2552/ae89e8

[2] Ming Chang, **Shuntaro Suzuki**, Takahiro Kurose, Takuya Ibaraki. "Pretraining alpha rhythm enhancement by neurofeedback facilitates short-term perceptual learning and improves visual acuity by facilitated consolidation", *Frontiers in Neuroergonomics*, vol. 5, 1399578, 2024. doi: 10.3389/fnrgo.2024.1399578

Conference Proceedings

[3] **Shuntaro Suzuki***, Kento Tokura*, Daichi Yashima*, Kanon Amemiya*, Komei Sugiura, Shinnosuke Takamichi. "ELSA: Acoustic Event-Level Semantic Alignment for Fine-Grained Reference-Free Text-to-Audio Evaluation", *INTERSPEECH*, pp. 138-142, 2026. doi: 10.21437/Interspeech.2026-914 (* equal contribution)

[4] **Shuntaro Suzuki**, Shuitsu Koyama, Shinnosuke Hirano, Shunya Nagashima. "PENGUIN: General Vital Sign Reconstruction from PPG with Flow Matching State Space Model", *IEEE ICASSP*, pp. 6361-6365, 2026. doi: 10.1109/ICASSP55912.2026.11463201

[5] Shunya Nagashima, **Shuntaro Suzuki**, Shuitsu Koyama, Shinnosuke Hirano. "A Decomposition-Based State Space Model for Multivariate Time-Series Forecasting", *IEEE ICASSP*, pp. 1241-1245, 2026. doi: 10.1109/ICASSP55912.2026.11462651.

[6] Ching-Chih Sung*, **Shuntaro Suzuki***, Francis Pingfan Chien*, Komei Sugiura, Yu Tsao. "Condition-Invariant fMRI Decoding of Speech Intelligibility with Deep State Space Model", *IEEE ICASSP*, pp. 7402-7406, 2026. doi: 10.1109/ICASSP55912.2026.11461445 (* equal contribution)

[7] Daichi Yashima, Shuhei Kurita, Yusuke Oda, **Shuntaro Suzuki**, Seitaro Otsuki, Komei Sugiura. "AB-MAMBA: Multimodal Large Language Model with Aligned Hierarchical Bidirectional Scan for Efficient Video Captioning", *ICPR*, pp. 409-424, 2026. doi: 10.1007/978-3-032-31654-7_28

Preprints

[8] **Shuntaro Suzuki**, Shunya Nagashima, Komei Sugiura. "BRAID: A Generalizable EEG Tokenizer via Acquisition-Aware Disentangled Representation Learning", 2026.

[9] **Shuntaro Suzuki**, Yuiga Wada, Komei Sugiura. "CANDLE: Cortical Null-Space Decomposition for Non-invasive Brain Source Imaging", 2026.

Honors and Awards

Funai Overseas Scholarship **Sep. 2027 – Aug. 2030**

- Up to **USD $150,000** over three years: $36,000/year living stipend, up to $14,000/year tuition support, and full health insurance.
- Awarded only for top 16 Japanese students applying for Ph.D. study abroad.

The 2025 PNPL Competition, NeurIPS, 2025 **Aug. 2025 – Sep. 2025**

- **Achieved 2nd place** in the Phoneme Classification Task, Standard Track (USD $1,000 award).
- Developed a deep learning model for phoneme decoding from SQUID-MEG signals recorded during naturalistic audiobook listening.

Research Experience

Semantic Machine Intelligence Lab, Keio University, Japan **Oct. 2023 – Present**
Advisor: Prof. Komei Sugiura

- Developed interpretable deep learning models for motor imagery decoding from ECoG/EEG signals to support communication for patients with ALS, in collaboration with a medical school.
- Developed a cortical-geometry-aware electrophysiological source imaging model trained solely on simulated data, achieving zero-shot transfer to intracranial stimulation localization and epileptogenic zone estimation.
- Resulting work includes first-author papers accepted in J. Neural Eng. [1] and under review [8], [9].

Suntory Global Innovation Center Limited, Japan **Jan. 2025 – Present**
Advisor: Dr. Haruo Mizutani, Prof. Akane Sano, Dr. Masaki Nakanishi

- Developed machine learning pipelines for circadian rhythm analysis using multimodal physiological signals (ECG, PPG, and EEG) acquired from an in-house wearable device.
- Developed an automated artifact detection and removal model for ECG signals using expert annotations.

Neurogica Inc., Japan **Jun. 2025 – Present**

- Developed models for reconstructing diverse physiological indicators (e.g., ECG, ABP, Respiratory Rate) from PPG signals, alongside general-purpose architectures for time-series modeling.
- This work resulted in one first-author and one second-author paper accepted to IEEE ICASSP 2026 [4], [5].

Biomedical Acoustic Signal Processing Lab, Academia Sinica, Taiwan **Aug. 2025 – Sep. 2025**
Advisor: Dr. Yu Tsao

- Developed a deep learning model for speech intelligibility decoding from fMRI signals recorded during speech listening under diverse acoustic conditions, leveraging an extended SSM for efficient fMRI modeling.
- Resulting work accepted at IEEE ICASSP 2026 as co-first author [6].

NTT DATA Institute of Management Consulting Inc., Japan **May 2022 – Oct. 2024**
Advisor: Dr. Takuya Ibaraki

- Conducted neurofeedback intervention studies using a proprietary wearable EEG device, contributing to experimental design and execution, analysis pipeline development, and manuscript preparation.
- Resulting work published in the peer-reviewed journal Front. Neuroergo. [2].

Skills

Programming: Python, MATLAB, C/C++, Bash

Machine Learning: Pytorch, Scikit-learn, pandas, Polars, NumPy

Neuroimaging: MNE-Python, Nilearn, NiBabel, EEGLAB

Tools: Git, GitHub/GitLab, Docker, SLURM/HPC

Languages: Japanese (Native), English (TOEFL iBT: 5.0/6.0, MyBest Score: 5.5/6.0)